Solving one-step equations

1 Determine whether the given value of x is a solution to the equation.

a $x = 6\ ;\ 8x = 51$

b $x = 4;\ 2x = 2 \times 4$

c $x = 19;\ \frac{x}{2} = 8$

d $x = 24;\ \frac{x}{3} = 8$

e $x = 9\ ;\ x - 10 = -1$

f $x = 0;\ x - 4 = -5$

g $x = 6\text{: } x + 1 = 7$

h $x = 3;\ x - 3 = -3$

2 Solve the following equations:

a $21 = x + 13$

b $x - 4 = 10$

c $7 = x - 1$

d $x - 8 = -20$

e $x + 12 = 9$

f $6 = -16 + n$

g $n + 27 = 11$

h $x - 21 = -25$

i $x - 3 = -5$

j $x - 1.39 = 8.67$

k $x - \frac{2}{5} = \frac{2}{5}$

l $x + \frac{1}{5} = \frac{3}{5}$

m $x - \frac{5}{8} = \frac{1}{8}$

n $x - \frac{3}{4} = \frac{1}{8}$

o $t - \frac{5}{8} = \frac{5}{7}$

3 Solve the following equations:

a $5x = 45$

b $12 = 3x$

c $3x = 18$

d $4x = -28$

e $-5x = 30$

f $-7x = 0$

g $-7x = -28$

h $-x = 3$

i $2 = -x$

j $-10x = -50$

k $\frac{x}{8} = 6$

l $\frac{1}{8}x = 6$

m $-\frac{1}{8}x = 6$

n $-5 = \frac{x}{5}$

o $-6 = \frac{y}{-7}$

p $\frac{1}{8}x = -8$

q $-\frac{1}{8}x = -8$

r $9t = 3$

s $\frac{t}{6} = 10$

t $88 = 11 \times n$

u $4 \times n = 27.2$

v $n \times 5.9 = 177$

w $36.5 = 7.3 \times n$

x $\frac{4}{9} \times n = 10$

y $\frac{y}{-1 \times 5} = -5$

4 For each of the following one-step equations, describe the step required to solve it:

a $x - 5 = 4$

b $x - 4.5 = 6.8$

c $9x = 7 + 8x$

d $\frac{x}{4.1} = 7.6$

e $4.1x = 94.3$

5 Find the number for each scenario:

a When 10 is added to a number, the result is 12.

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b When a number is multiplied by 10, the result is 60.

Solving two-step equations

6 Solve the following equations:

a $7x + 14 = 0$

b $-x + 5 = -1$

c $-x - 7 = 7$

d $\frac{x}{2} + 8 = 10$

e $\frac{x}{7} + 3 = -1$

f $4x - 6x = 4$

g $-6x + 4x - 3x = 0$

h $-63 - 9y = 63$

i $3 = \frac{u}{4} - 5$

j $7 = \frac{m + 10}{5}$

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